Tyler Cartwright

Machine Learning Engineer — fraud & economic crime systems

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I build the systems that catch financial crime — machine learning that finds the handful of transactions that matter, hidden in millions that don't. Four roles at Lloyds Banking Group, one arc: infrastructure → data → models → production ML.

EXPERIENCE — LLOYDS BANKING GROUP

Machine Learning Engineer · Economic Crime Prevention Platform

Jan 2026 – Present

+ First MLE on the team: own the shared Python utilities package, standardising workflows across model sub-teams and eliminating duplicated effort
+ Productionising fraud detection models on GCP Vertex AI — translating notebook experiments into auditable, scalable batch-inference pipelines

Data Scientist · Economic Crime Prevention Platform

May 2025 – Jan 2026

+ Built two machine learning models for AML and fraud detection
+ Improved detection rates by 50% per model without increasing alert volume
+ Each model recovers an additional £100,000 per month

Data Engineer · Economic Crime Prevention Platform

Sept 2024 – May 2025

+ Designed and ran Spark ETL pipelines processing millions of transactions daily
+ Introduced 100+ new data points to the warehouse, directly improving model accuracy
+ Optimised Spark applications for performance and cost, saving £500,000 per year

Cloud Software Engineer (Placement) · Homes Platform

June 2022 – Sept 2023

+ Built incident detection in Azure, cutting response times from hours to minutes
+ Migrated a central microservices application to Azure; automated health checks and alerting, halving downtime

FREELANCE & PROJECTS

Staff Management System · Hodsock Priory & Prestwold Hall

June 2024 – May 2025

+ Full-stack shift-tracking and payroll platform (React/Node.js) with role-based access and automated reporting — cut administrative overhead by 95%

WaniKani Statistics Tracker · Open source

+ TypeScript PWA visualising learning progress, used by thousands of Japanese learners

EDUCATION

BSc Computer Science — First Class Honours · University of Derby

2020 – 2024

+ Dean's Award recipient every year; specialised in Machine Learning and Data Science
+ Final-year project: sentiment analysis of Reddit data during the GameStop short squeeze

SKILLS

Languages	Python · SQL · JavaScript · TypeScript · Java
Cloud & MLOps	GCP · Vertex AI · Azure · Docker · Kubernetes · Terraform
Data & ML	Apache Spark · BigQuery · Scikit-learn · LightGBM · PyTorch · ETL pipelines
Web & practices	React · Node.js · REST APIs · Git · CI/CD · GitHub Actions · Agile/Scrum